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Roll-No:

2024-CET-04

Department:

Software Engineering

Group:

Computer Engineering Technology

Semester:

4th Semester

Subject:

Database Applications

Lab Task:

SQL Server – Keys and Constraints

Submitted To:

Engr. Asma Javid

Lab Task 8

Section 1:

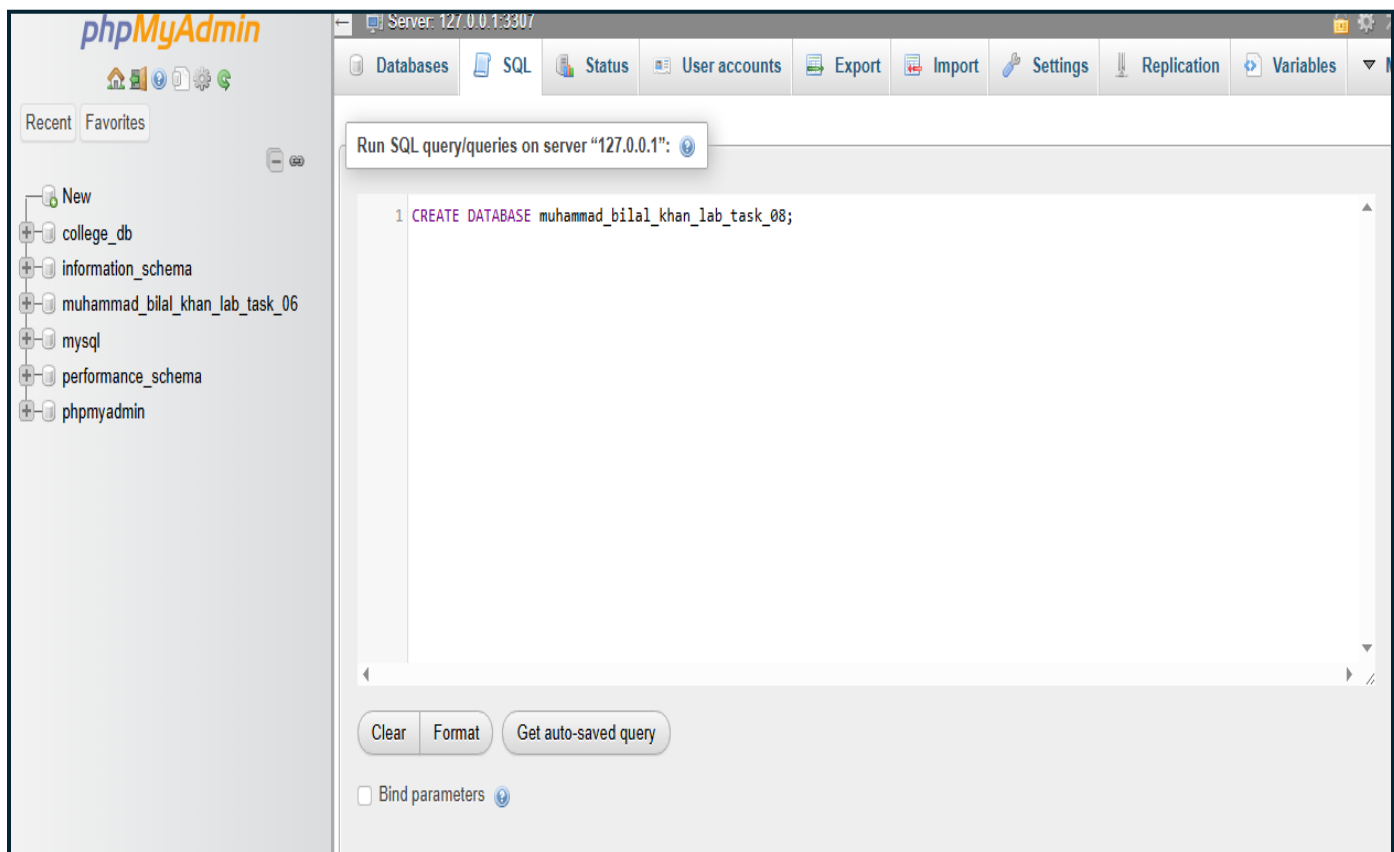
Creating Database and Tables with Initial Constraints

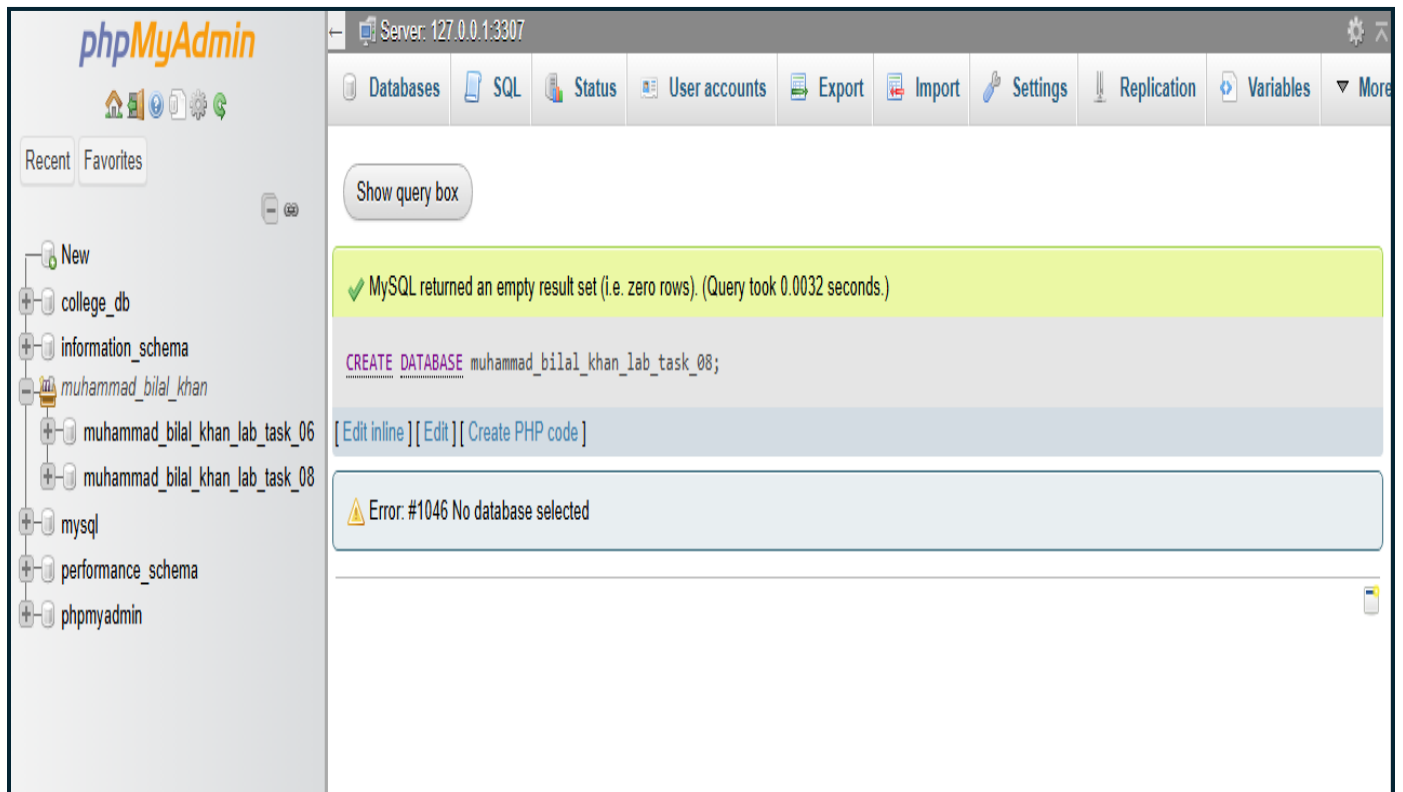
create a database and define tables with primary key, NOT NULL, and default constraints.

1.Create a new database named as your full name in all lowercase with underscores,

followed by _lab_task_08.

Example: kainat_pervaiz_lab_task_08





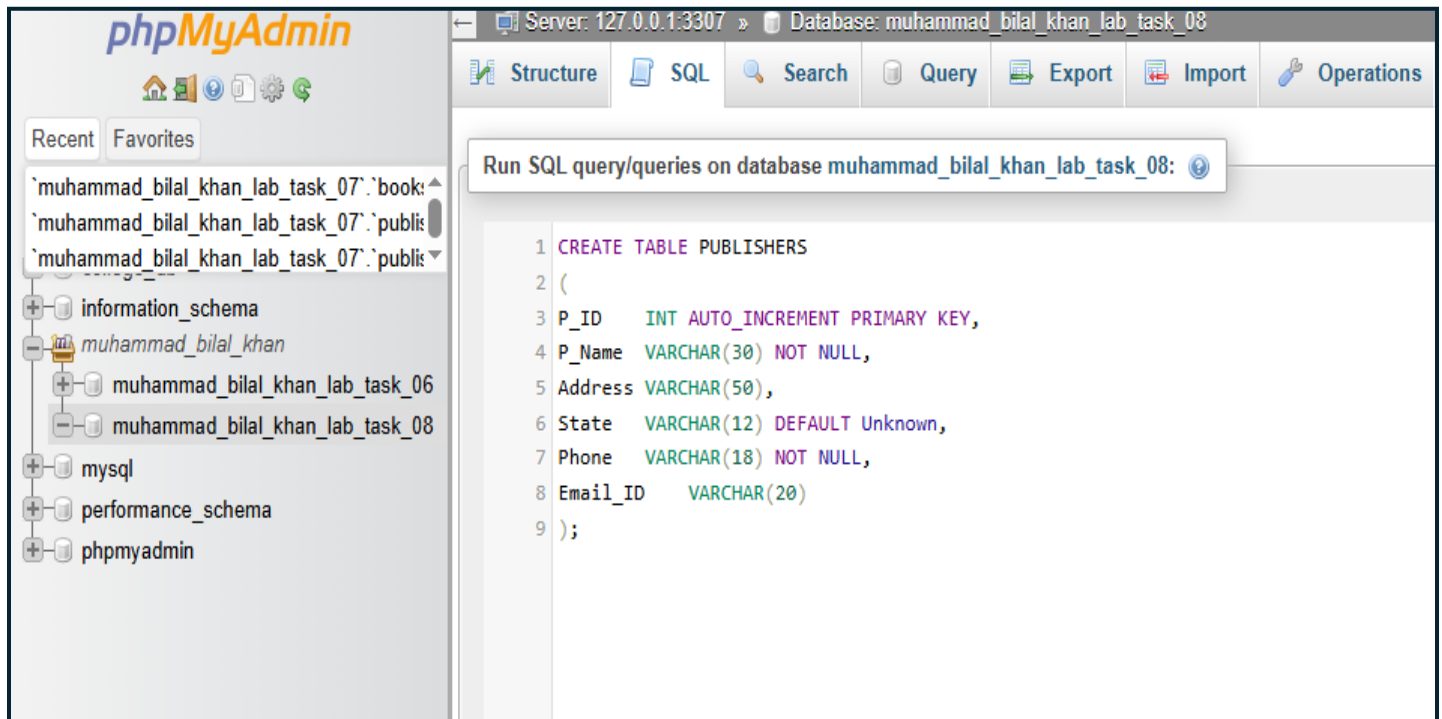
2. Inside this database, create a table named PUBLISHERS with the following structure:

Attribute	Data Type	Field Size
P_ID	INT	---
P_Name	VARCHAR	30
Address	VARCHAR	50
State	VARCHAR	12
Phone	VARCHAR	18
Email_ID	VARCHAR	20

Set P_ID as the Primary Key while creating the table.

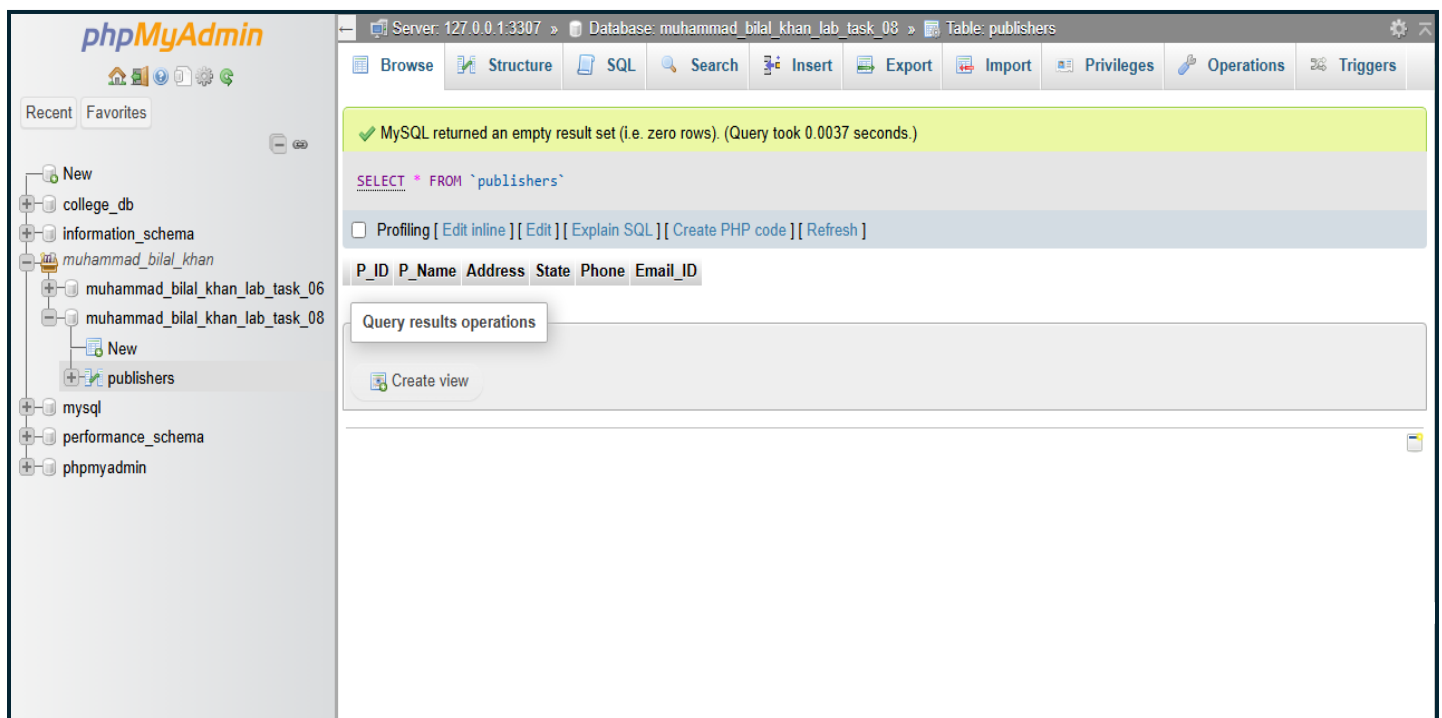
Make sure that P_Name and Phone are NOT NULL while creating the table.

Set a DEFAULT value of 'Unknown' for the State attribute.



The screenshot shows the phpMyAdmin interface with the SQL editor open. The database selected is 'muhammad_bilal_khan_lab_task_08'. The SQL query being entered is:

```
1 CREATE TABLE PUBLISHERS
2 (
3   P_ID INT AUTO_INCREMENT PRIMARY KEY,
4   P_Name VARCHAR(30) NOT NULL,
5   Address VARCHAR(50),
6   State VARCHAR(12) DEFAULT Unknown,
7   Phone VARCHAR(18) NOT NULL,
8   Email_ID VARCHAR(20)
9 );
```



The screenshot shows the phpMyAdmin interface with the 'publishers' table selected. The table structure is displayed, and a query has been executed, resulting in an empty result set.

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0037 seconds.)

`SELECT * FROM `publishers``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

P_ID	P_Name	Address	State	Phone	Email_ID
------	--------	---------	-------	-------	----------

Query results operations

[Create view](#)

Section 02:

Applying More Constraints Using ALTER TABLE

modify table structure by adding additional constraints using ALTER TABLE.

1. Create a table named BOOKS with the following structure:

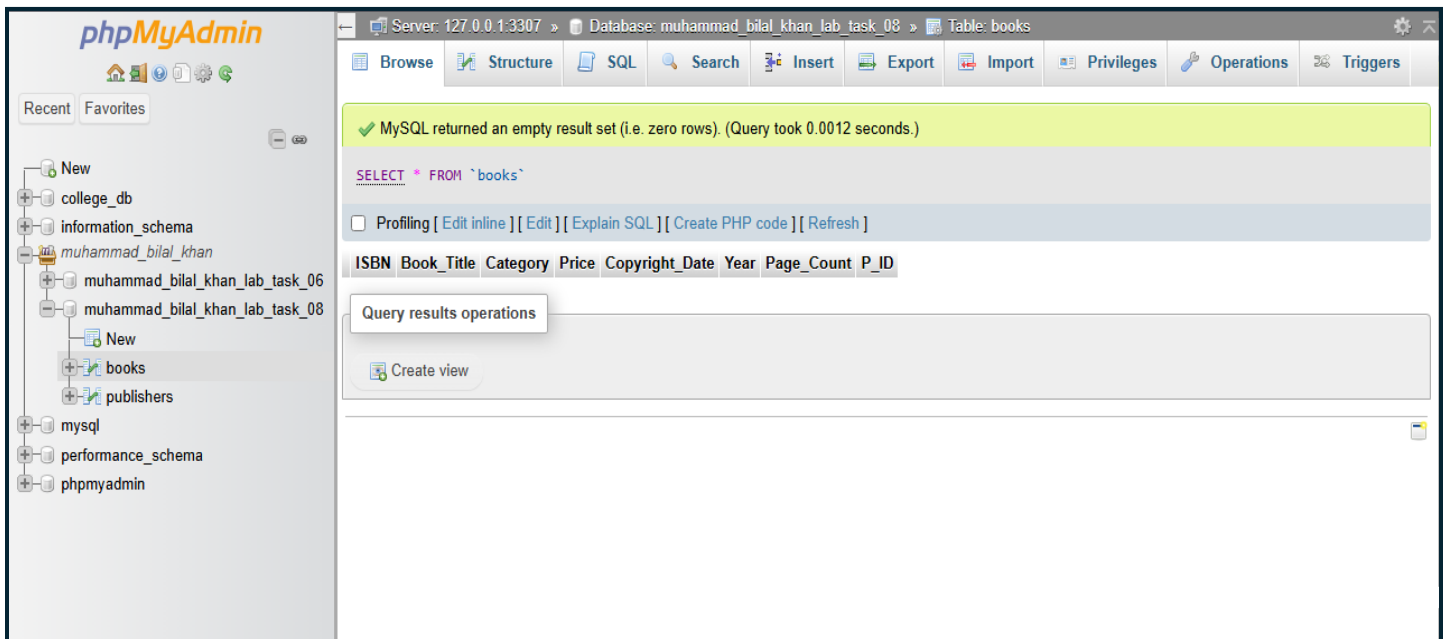
Attribute	Data Type	Field Size
ISBN	INT	---
Book_Title	VARCHAR	50
Category	VARCHAR	20
Price	NUMERIC	(6,2)
Copyright_Date	VARCHAR	4
Year	NUMERIC	4
Page_Count	NUMERIC	4
P_ID	INT	---

Set ISBN as the Primary Key while creating the table.

Set Book_Title and P_ID as NOT NULL fields during creation.

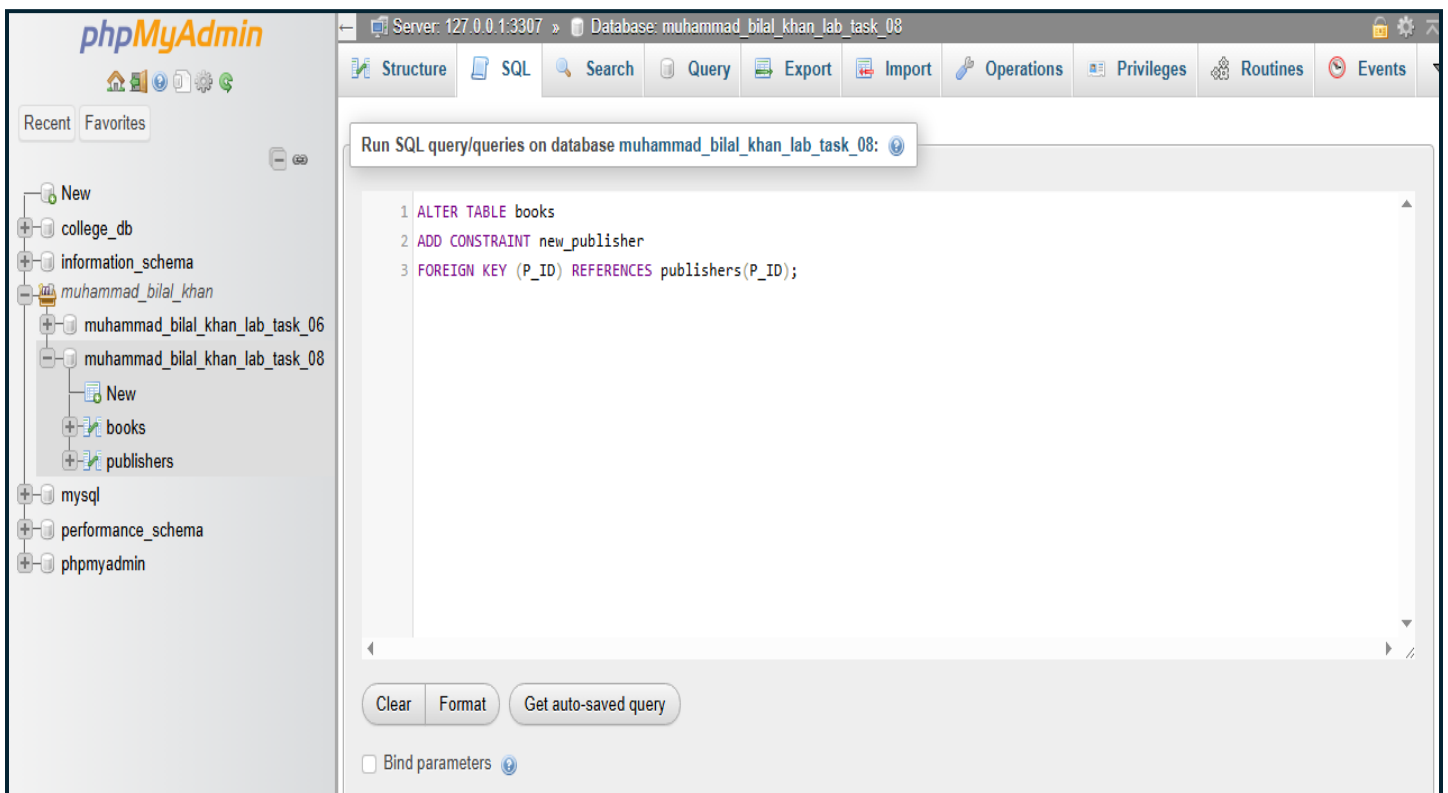
The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including a database named 'muhammad_bilal_khan_lab_task_08'. The main panel displays the SQL query to create the 'BOOKS' table:

```
1 CREATE TABLE BOOKS
2 (
3 ISBN          INT AUTO_INCREMENT PRIMARY KEY,
4 Book_Title    VARCHAR(50) NOT NULL,
5 Category      VARCHAR(20),
6 Price         NUMERIC(6,2),
7 Copyright_Date VARCHAR(4),
8 Year          NUMERIC(4),
9 Page_Count    NUMERIC(4),
10 P_ID          INT NOT NULL
11 );
```



2. After creating the BOOKS table:

Add a Foreign Key on P_ID referencing P_ID from the PUBLISHERS table.



Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: publishers

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)

```
SELECT * FROM `publishers`
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

P_ID	P_Name	Address	State	Phone	Email_ID
------	--------	---------	-------	-------	----------

Query results operations

[Create view](#)

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: books

[Table structure](#) [Relation view](#)

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	ISBN	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐ 2	Book_Title	varchar(50)	utf8mb4_general_ci		No	None			Change Drop More
☐ 3	Category	varchar(20)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 4	Price	decimal(6,2)			Yes	NULL			Change Drop More
☐ 5	Copyright_Date	varchar(4)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐ 6	Year	decimal(4,0)			Yes	NULL			Change Drop More
☐ 7	Page_Count	decimal(4,0)			Yes	NULL			Change Drop More
☐ 8	P_ID	int(11)			No	None			Change Drop More

☐ Check all With selected: [Browse](#) [Change](#) [Drop](#) [Primary](#) [Unique](#) [Index](#) [Spatial](#)

[Fulltext](#)

[Print](#) [Propose table structure](#) [Move columns](#) [Normalize](#)

[Add](#) 1 column(s) after P_ID [Go](#)

[Indexes](#)

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	ISBN	0	A	No	
Edit Rename Drop	new_publisher	BTREE	No	No	P_ID	0	A	No	

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
- muhammad_bilal_khan_lab_task_06
- muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: books

Structure SQL Search Insert Export Import Privileges Operations Triggers

Table structure Relation view

Foreign key constraints

Actions	Constraint properties	Column	Foreign key constraint (INNODB)		
			Database	Table	Column
Drop	ON DELETE RESTRICT	P_ID	muhammad_bilal_khan	publishers	P_ID
	ON UPDATE RESTRICT				
	Constraint name				
	ON DELETE RESTRICT		muhammad_bilal_khan		
	ON UPDATE RESTRICT				

+ Add constraint

Preview SQL Save

Add a CHECK Constraint to ensure Price is greater than 20.

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
- muhammad_bilal_khan_lab_task_06
- muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08

Structure SQL Search Query Export Import Operations

Run SQL query/queries on database muhammad_bilal_khan_lab_task_08:

```

1 ALTER TABLE BOOKS
2 ADD CONSTRAINT chk_price,
3 CHECK (Price > 20)
  
```

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
- muhammad_bilal_khan_lab_task_06
- muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: books

Browse Structure SQL Search Insert Export Import Privileges

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0753 seconds.)

```
ALTER TABLE BOOKS ADD CONSTRAINT chk_price CHECK (Price > 20);
```

[Edit inline] [Edit] [Create PHP code]

Add a UNIQUE Constraint on Book_Title to ensure no two books have the same title.

The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'college_db', 'information_schema', 'muhammad_bilal_khan', and 'muhammad_bilal_khan_lab_task_08'. The 'books' table is selected under 'muhammad_bilal_khan_lab_task_08'. The main panel shows the 'SQL' tab with the following query:

```
1 ALTER TABLE books
2 ADD CONSTRAINT uq_book_title
3 UNIQUE (Book_Title);
```

Buttons for 'Clear', 'Format', and 'Get auto-saved query' are visible at the bottom of the query editor.

The screenshot shows the phpMyAdmin interface after the query has been executed. A green message box states: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0182 seconds.)'. Below the message, the executed SQL query is displayed: 'ALTER TABLE books ADD CONSTRAINT uq_book_title UNIQUE (Book_Title);'. Links for '[Edit inline]', '[Edit]', and '[Create PHP code]' are provided.

The screenshot shows the phpMyAdmin interface with the 'Structure' tab selected for the 'books' table. The table structure is as follows:

#	Field Name	Type	Charset	Collation	Nullable	Index
2	Book_Title	varchar(50)	utf8mb4_general_ci		No	None
3	Category	varchar(20)	utf8mb4_general_ci		Yes	NULL
4	Price	decimal(6,2)			Yes	NULL
5	Copyright_Date	varchar(4)	utf8mb4_general_ci		Yes	NULL
6	Year	decimal(4,0)			Yes	NULL
7	Page_Count	decimal(4,0)			Yes	NULL
8	P_ID	int(11)			No	None

Below the table structure, the 'Indexes' section is expanded, showing the following indexes:

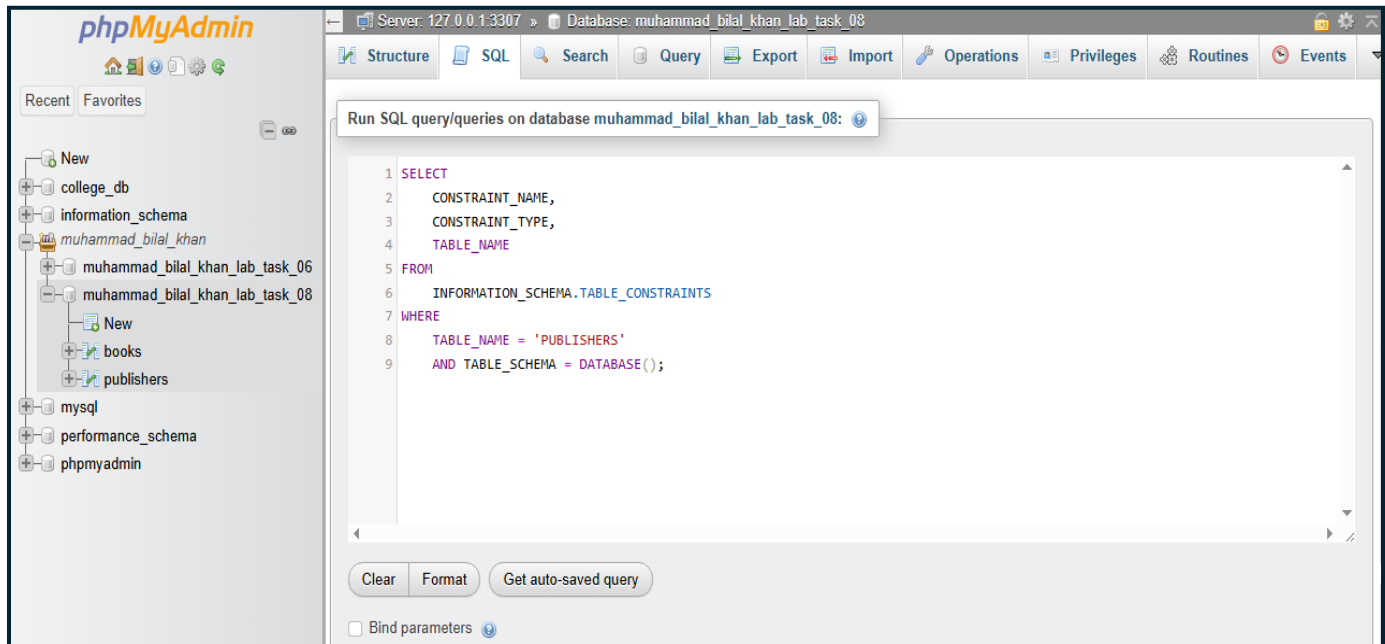
Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit	PRIMARY	BTREE	Yes	No	ISBN	0	A	No	
Edit	uq_book_title	BTREE	Yes	No	Book_Title	0	A	No	
Edit	new_publisher	BTREE	No	No	P_ID	0	A	No	

Section 03:

Viewing and Describing Table Constraints

how to view existing constraints and understand their structure.

1.To view the list of constraints applied to both PUBLISHERS and BOOKS tables.



The screenshot shows the phpMyAdmin interface. On the left, the database structure tree is visible, with 'muhammad_bilal_khan_lab_task_08' selected. The main panel displays a SQL query editor with the following query:

```
1 SELECT
2   CONSTRAINT_NAME,
3   CONSTRAINT_TYPE,
4   TABLE_NAME
5 FROM
6   INFORMATION_SCHEMA.TABLE_CONSTRAINTS
7 WHERE
8   TABLE_NAME = 'PUBLISHERS'
9   AND TABLE_SCHEMA = DATABASE();
```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present.



The screenshot shows the phpMyAdmin interface after executing the query. A green message bar indicates: "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)". Below this, the SQL query is repeated. A 'Profiling' section is visible with links for 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A table header is shown:

CONSTRAINT_NAME	CONSTRAINT_TYPE	TABLE_NAME
-----------------	-----------------	------------

Below the table header, there is a 'Query results operations' section with a 'Create view' button.

phpMyAdmin

Recent Favorites

New

- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08

Structure SQL Search Query Export Import Operations

Run SQL query/queries on database muhammad_bilal_khan_lab_task_08:

```
1 SELECT
2     CONSTRAINT_NAME,
3     CONSTRAINT_TYPE,
4     TABLE_NAME
5 FROM
6     INFORMATION_SCHEMA.TABLE_CONSTRAINTS
7 WHERE
8     TABLE_NAME = 'BOOKS'
9     AND TABLE_SCHEMA = DATABASE();
```

phpMyAdmin

Recent Favorites

New

- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: TABLE_CONSTRAINTS

Browse Structure SQL Search Insert Export Import Privileges

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds.)

```
SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE, TABLE_NAME FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS WHERE
TABLE_SCHEMA = DATABASE();
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

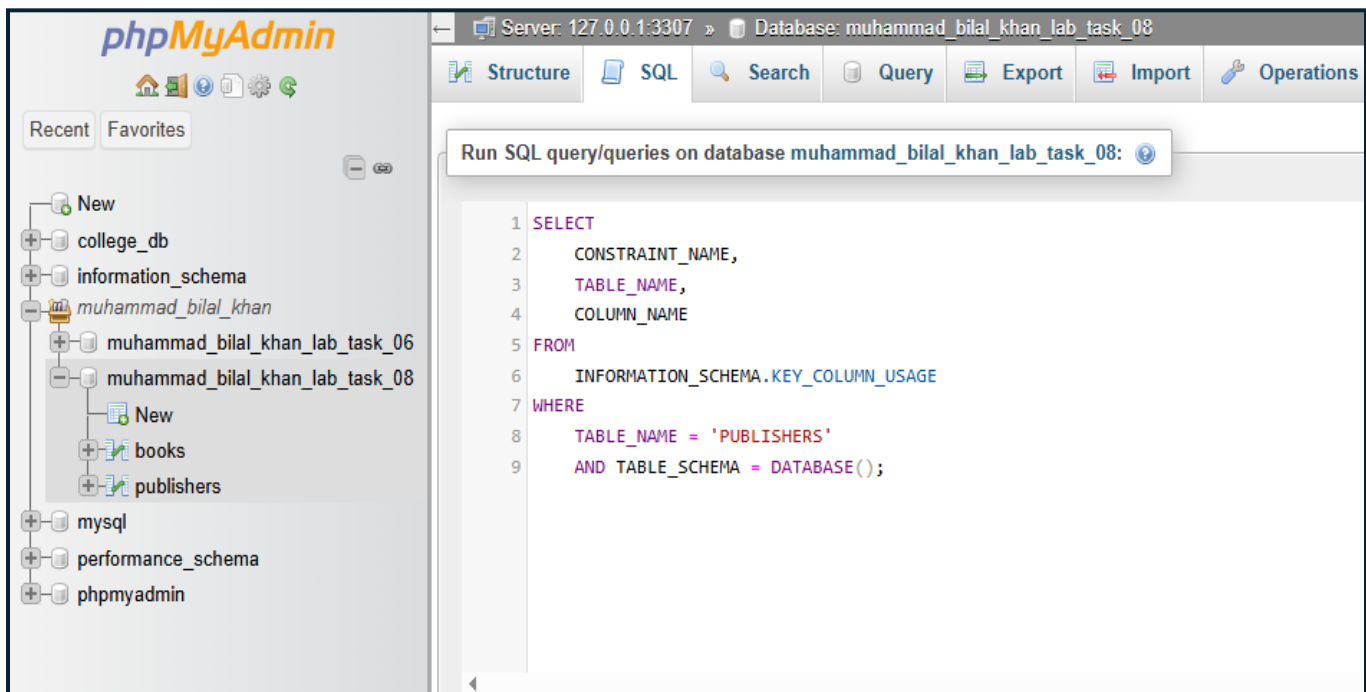
CONSTRAINT_NAME	CONSTRAINT_TYPE	TABLE_NAME
-----------------	-----------------	------------

Query results operations

Create view

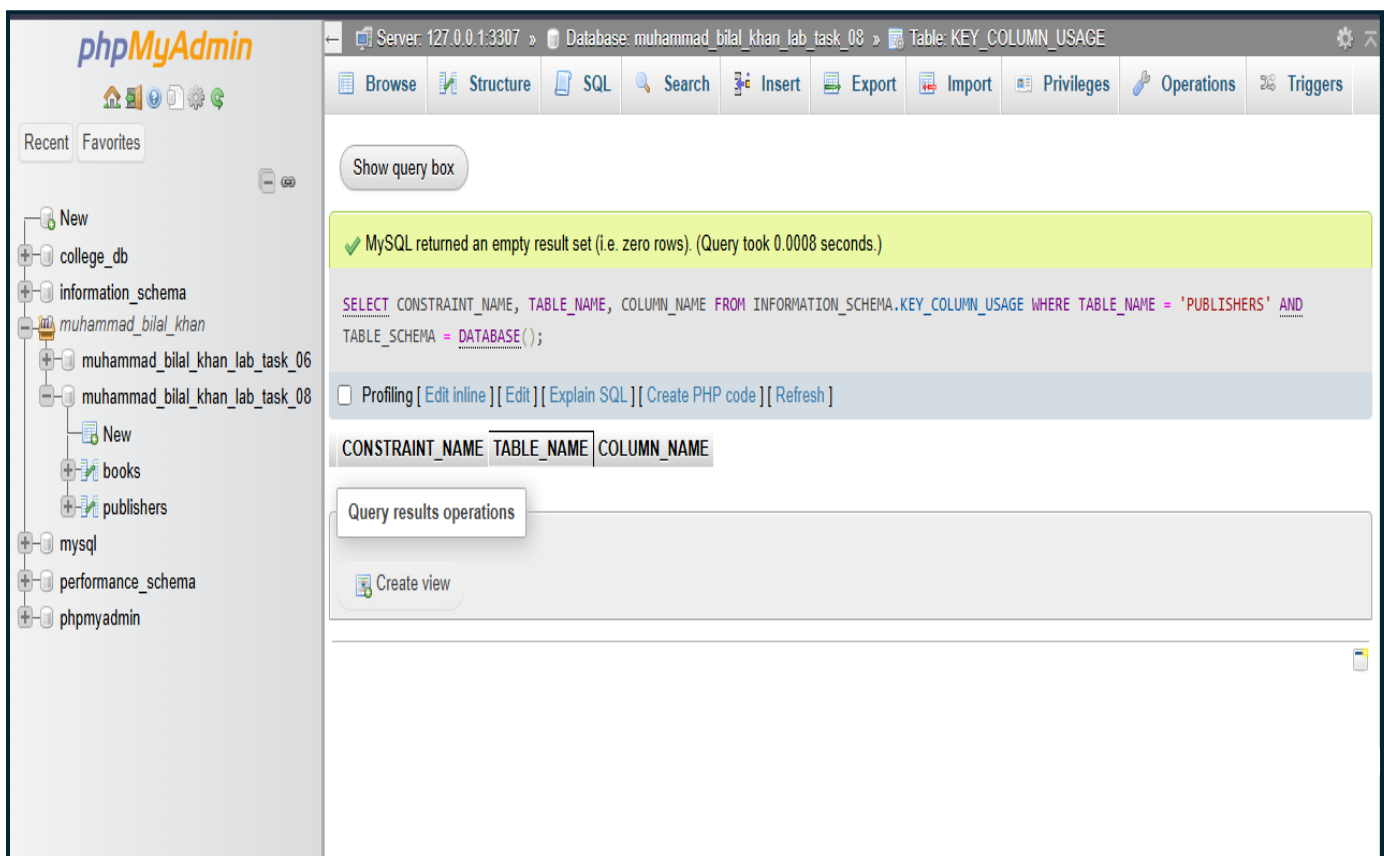
2. Specifically, query:

INFORMATION_SCHEMA.TABLE_CONSTRAINTS



The screenshot shows the phpMyAdmin interface with the SQL tab selected. The database 'muhammad_bilal_khan_lab_task_08' is selected. The SQL query editor contains the following query:

```
1 SELECT
2     CONSTRAINT_NAME,
3     TABLE_NAME,
4     COLUMN_NAME
5 FROM
6     INFORMATION_SCHEMA.KEY_COLUMN_USAGE
7 WHERE
8     TABLE_NAME = 'PUBLISHERS'
9     AND TABLE_SCHEMA = DATABASE();
```



The screenshot shows the phpMyAdmin interface with the query results displayed. The message indicates that the query returned an empty result set.

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)

The query executed was:

```
SELECT CONSTRAINT_NAME, TABLE_NAME, COLUMN_NAME FROM INFORMATION_SCHEMA.KEY_COLUMN_USAGE WHERE TABLE_NAME = 'PUBLISHERS' AND TABLE_SCHEMA = DATABASE();
```

Below the message, there are links for Profiling, Edit inline, Edit, Explain SQL, Create PHP code, and Refresh. A table header is visible with columns: CONSTRAINT_NAME, TABLE_NAME, and COLUMN_NAME. Below the table header, there is a section for Query results operations, including a Create view button.

INFORMATION_SCHEMA.CONSTRAINT_COLUMN_USAGE

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: KEY_COLUMN_USAGE

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_08.KEY_COLUMN_USAGE:

```

1 SELECT
2     CONSTRAINT_NAME,
3     TABLE_NAME,
4     COLUMN_NAME
5 FROM
6     INFORMATION_SCHEMA.KEY_COLUMN_USAGE
7 WHERE
8     TABLE_NAME = 'BOOKS'
9     AND TABLE_SCHEMA = DATABASE();

```

Clear Format Get auto-saved query

☐ Bind parameters

phpMyAdmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: KEY_COLUMN_USAGE

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0019 seconds.)

```

SELECT CONSTRAINT_NAME, TABLE_NAME, COLUMN_NAME FROM INFORMATION_SCHEMA.KEY_COLUMN_USAGE WHERE TABLE_NAME = 'BOOKS' AND
TABLE_SCHEMA = DATABASE();

```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

CONSTRAINT_NAME	TABLE_NAME	COLUMN_NAME
-----------------	------------	-------------

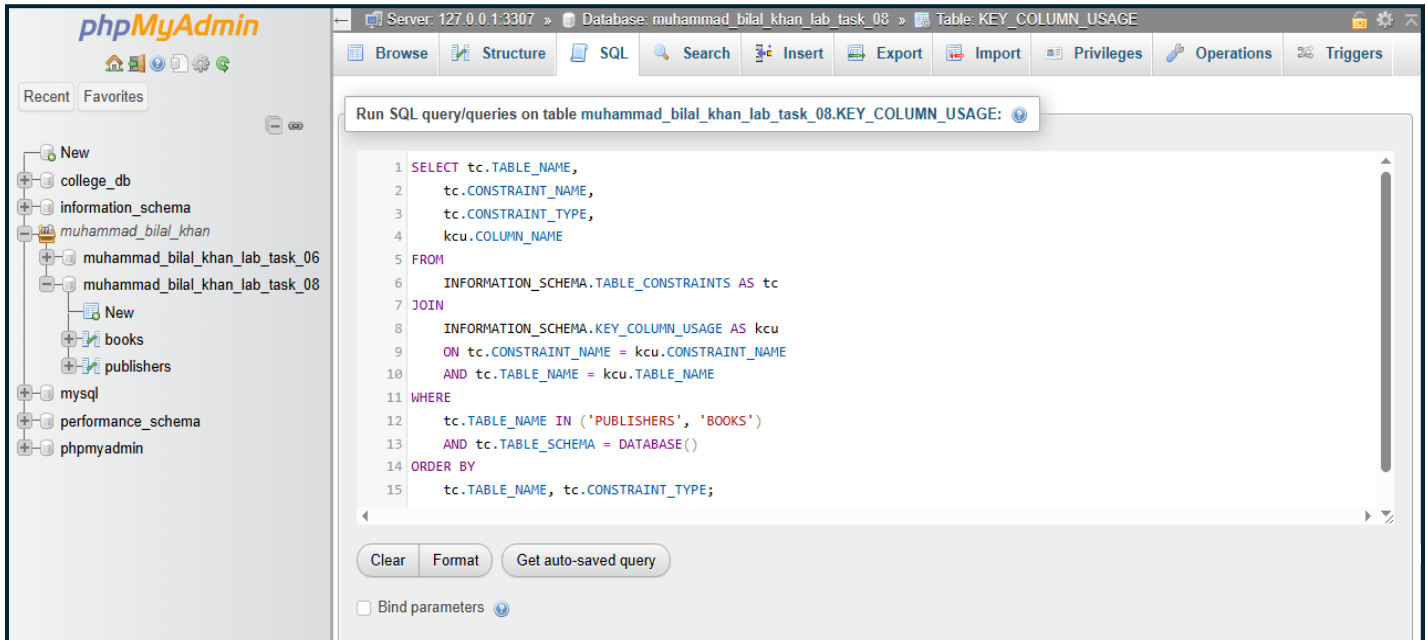
Query results operations

Create view

3. Write queries that:

Display the names of all constraints on each table.

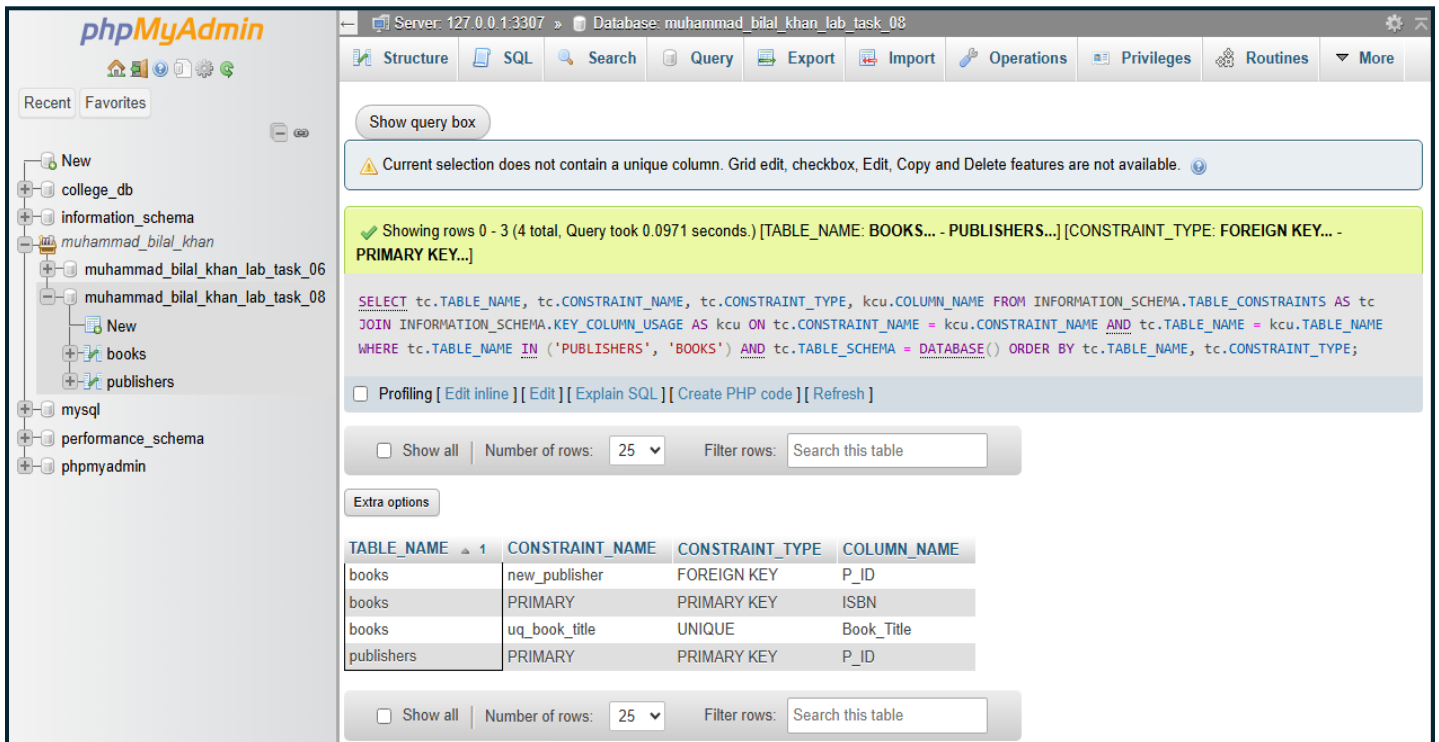
Display which columns are associated with which constraints.



The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including 'college_db', 'information_schema', 'muhammad_bilal_khan', and 'muhammad_bilal_khan_lab_task_08'. The main panel displays a SQL query to find columns associated with constraints for the 'PUBLISHERS' and 'BOOKS' tables.

```
1 SELECT tc.TABLE_NAME,
2       tc.CONSTRAINT_NAME,
3       tc.CONSTRAINT_TYPE,
4       kcu.COLUMN_NAME
5 FROM
6   INFORMATION_SCHEMA.TABLE_CONSTRAINTS AS tc
7 JOIN
8   INFORMATION_SCHEMA.KEY_COLUMN_USAGE AS kcu
9   ON tc.CONSTRAINT_NAME = kcu.CONSTRAINT_NAME
10  AND tc.TABLE_NAME = kcu.TABLE_NAME
11 WHERE
12   tc.TABLE_NAME IN ('PUBLISHERS', 'BOOKS')
13   AND tc.TABLE_SCHEMA = DATABASE()
14 ORDER BY
15   tc.TABLE_NAME, tc.CONSTRAINT_TYPE;
```

4. Document your findings.



The screenshot shows the phpMyAdmin interface displaying the results of the SQL query. The results are shown in a table with 4 rows and 4 columns: TABLE_NAME, CONSTRAINT_NAME, CONSTRAINT_TYPE, and COLUMN_NAME.

TABLE_NAME	CONSTRAINT_NAME	CONSTRAINT_TYPE	COLUMN_NAME
books	new_publisher	FOREIGN KEY	P_ID
books	PRIMARY	PRIMARY KEY	ISBN
books	uq_book_title	UNIQUE	Book_Title
publishers	PRIMARY	PRIMARY KEY	P_ID

Section 04:

Dropping Specific Constraints

1.Drop the DEFAULT constraint on the State column from the PUBLISHERS table.

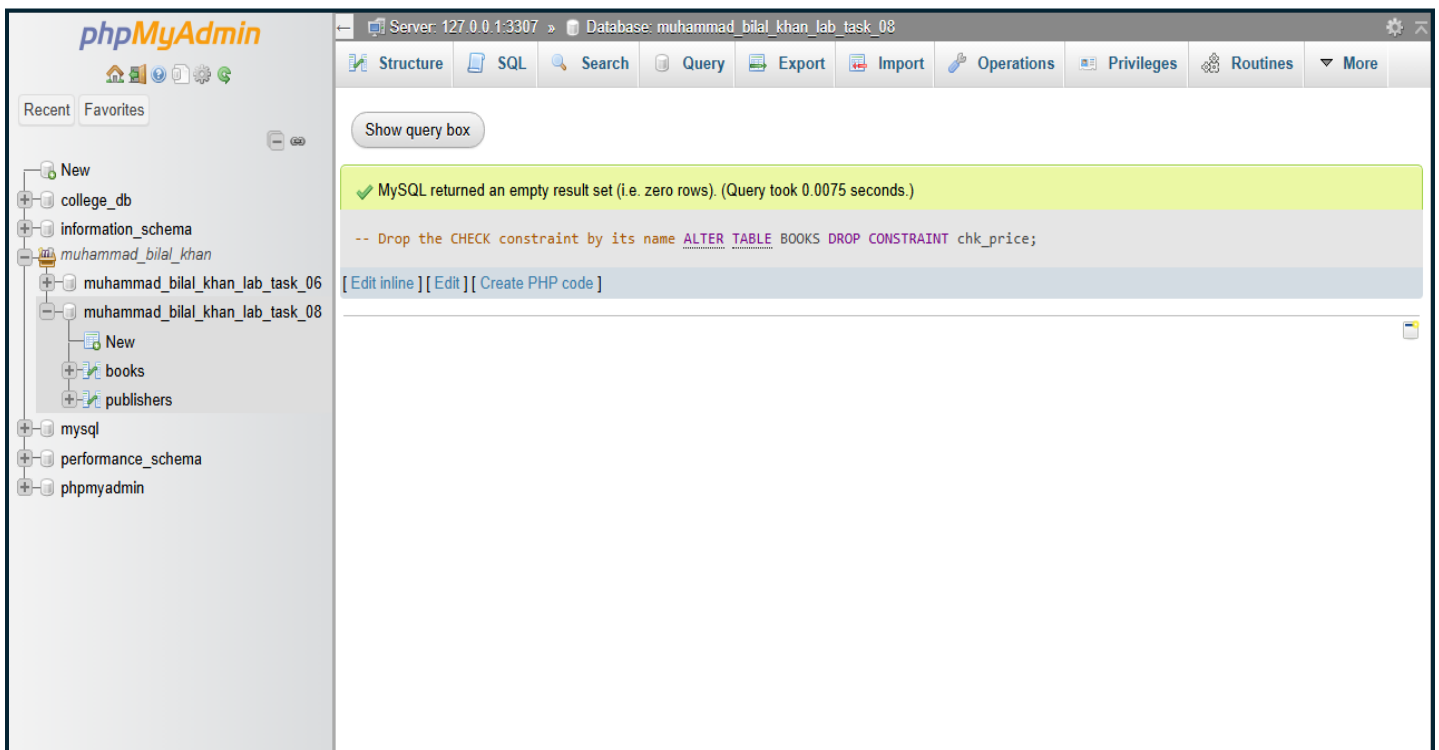
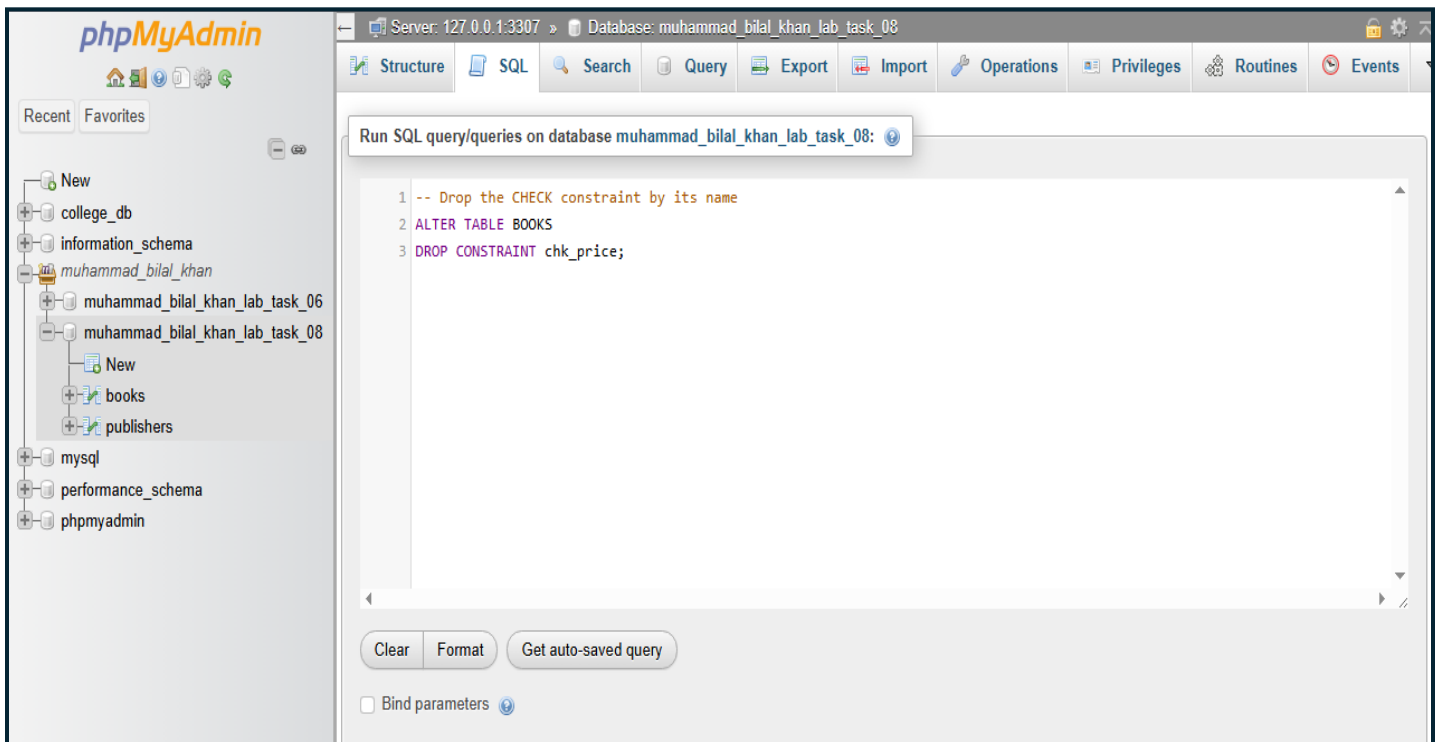
The screenshot shows the phpMyAdmin interface. On the left is the database navigation tree with 'muhammad_bilal_khan' selected and 'publishers' highlighted. The top toolbar includes 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', 'Privileges', 'Operations', and 'Triggers'. The 'SQL' tab is active, displaying a text area with the following SQL query:

```
1 -- Remove the DEFAULT value from State column
2 ALTER TABLE PUBLISHERS
3 MODIFY COLUMN State VARCHAR(50) NULL;
```

Below the text area are buttons for 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is also present.

This screenshot shows the same phpMyAdmin interface after the query has been executed. The 'Query' tab is now selected in the toolbar. A green message box states: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0606 seconds.)'. Below this, the executed SQL query is displayed: '-- Remove the DEFAULT value from State column ALTER TABLE PUBLISHERS MODIFY COLUMN State VARCHAR(50) NULL;'. At the bottom of the query area are links for '[Edit inline]', '[Edit]', and '[Create PHP code]'.

2.Drop the CHECK constraint applied to the Price column from the BOOKS table.



3. Make sure only the constraints are dropped, and the tables/data remain intact.

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08

Structure SQL Search Query Export Import Operations Privileges Routines Events

Run SQL query/queries on database muhammad_bilal_khan_lab_task_08:

```
1 -- Confirm PUBLISHERS table data is intact
2 SELECT * FROM PUBLISHERS;
3
4 -- Confirm BOOKS table data is intact
5 SELECT * FROM BOOKS;
```

Clear Format Get auto-saved query

☐ Bind parameters

phpMyAdmin

Recent Favorites

- New
- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: BOOKS

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0087 seconds.)

-- Confirm PUBLISHERS table data is intact `SELECT * FROM PUBLISHERS;`

[Edit inline] [Edit] [Create PHP code]

P_ID	P_Name	Address	State	Phone	Email_ID
------	--------	---------	-------	-------	----------

Query results operations

Create view

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0033 seconds.)

-- Confirm BOOKS table data is intact `SELECT * FROM BOOKS;`

[Edit inline] [Edit] [Create PHP code]

ISBN	Book Title	Category	Price	Copyright Date	Year	Page Count	P_ID
------	------------	----------	-------	----------------	------	------------	------

phpMyAdmin

Recent Favorites

New

- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: BOOKS

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on table muhammad_bilal_khan_lab_task_08.BOOKS:

```
1 SELECT tc.TABLE_NAME,
2     tc.CONSTRAINT_NAME,
3     tc.CONSTRAINT_TYPE,
4     kcu.COLUMN_NAME
5 FROM
6     INFORMATION_SCHEMA.TABLE_CONSTRAINTS AS tc
7 JOIN
8     INFORMATION_SCHEMA.KEY_COLUMN_USAGE AS kcu
9 ON tc.CONSTRAINT_NAME = kcu.CONSTRAINT_NAME
10 AND tc.TABLE_NAME = kcu.TABLE_NAME
11 WHERE
12     tc.TABLE_NAME IN ('PUBLISHERS', 'BOOKS')
13     AND tc.TABLE_SCHEMA = DATABASE()
14 ORDER BY
15     tc.TABLE_NAME, tc.CONSTRAINT_TYPE;
```

ISBN
Book_Title
Category
Price
Copyright_Date
Year
Page_Count
P_ID

SELECT * SELECT INSERT UPDATE DELETE Clear Format Get auto-saved query

☐ Bind parameters

phpMyAdmin

Recent Favorites

New

- college_db
- information_schema
- muhammad_bilal_khan
 - muhammad_bilal_khan_lab_task_06
 - muhammad_bilal_khan_lab_task_08
 - New
 - books
 - publishers
- mysql
- performance_schema
- phpmyadmin

Server: 127.0.0.1:3307 » Database: muhammad_bilal_khan_lab_task_08 » Table: BOOKS

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✔ Showing rows 0 - 3 (4 total, Query took 0.1247 seconds.) [TABLE_NAME: BOOKS... - PUBLISHERS...] [CONSTRAINT_TYPE: FOREIGN KEY... - PRIMARY KEY...]

```
SELECT tc.TABLE_NAME, tc.CONSTRAINT_NAME, tc.CONSTRAINT_TYPE, kcu.COLUMN_NAME FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS AS tc
JOIN INFORMATION_SCHEMA.KEY_COLUMN_USAGE AS kcu ON tc.CONSTRAINT_NAME = kcu.CONSTRAINT_NAME AND tc.TABLE_NAME = kcu.TABLE_NAME
WHERE tc.TABLE_NAME IN ('PUBLISHERS', 'BOOKS') AND tc.TABLE_SCHEMA = DATABASE() ORDER BY tc.TABLE_NAME, tc.CONSTRAINT_TYPE;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

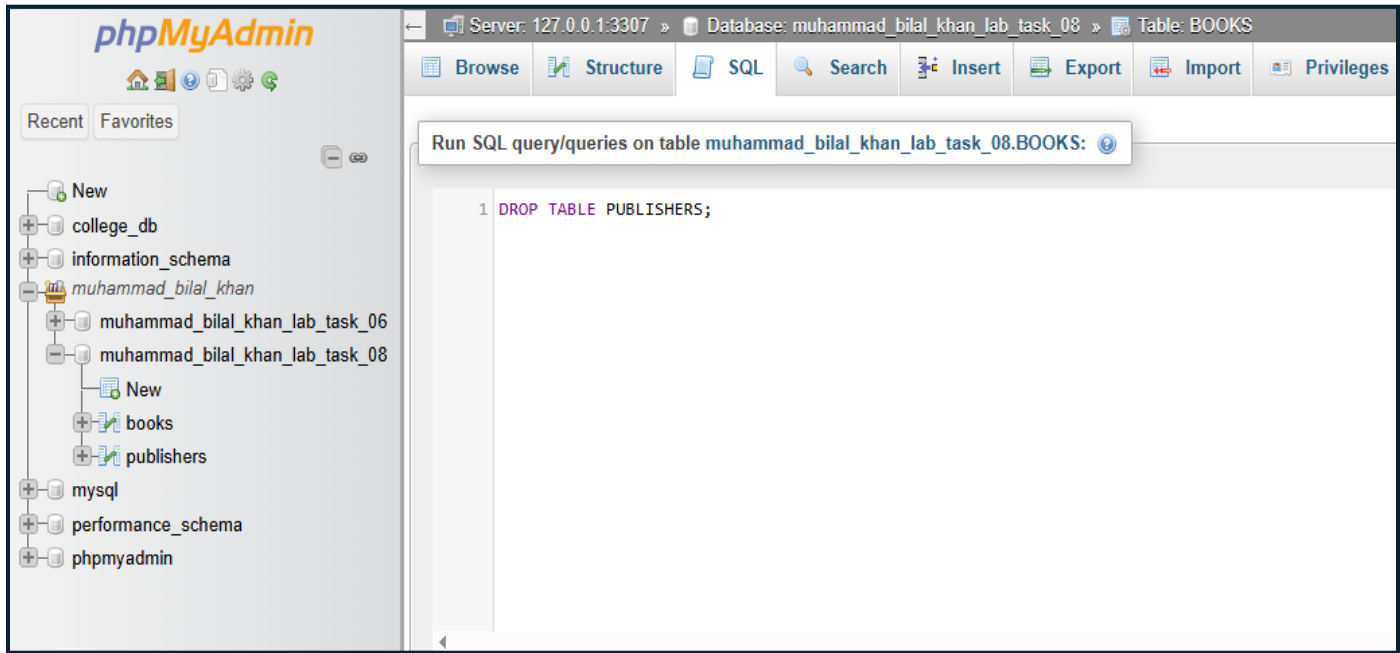
Extra options

TABLE_NAME	CONSTRAINT_NAME	CONSTRAINT_TYPE	COLUMN_NAME
books	new_publisher	FOREIGN KEY	P_ID
books	PRIMARY	PRIMARY KEY	ISBN
books	uq_book_title	UNIQUE	Book_Title
publishers	PRIMARY	PRIMARY KEY	P_ID

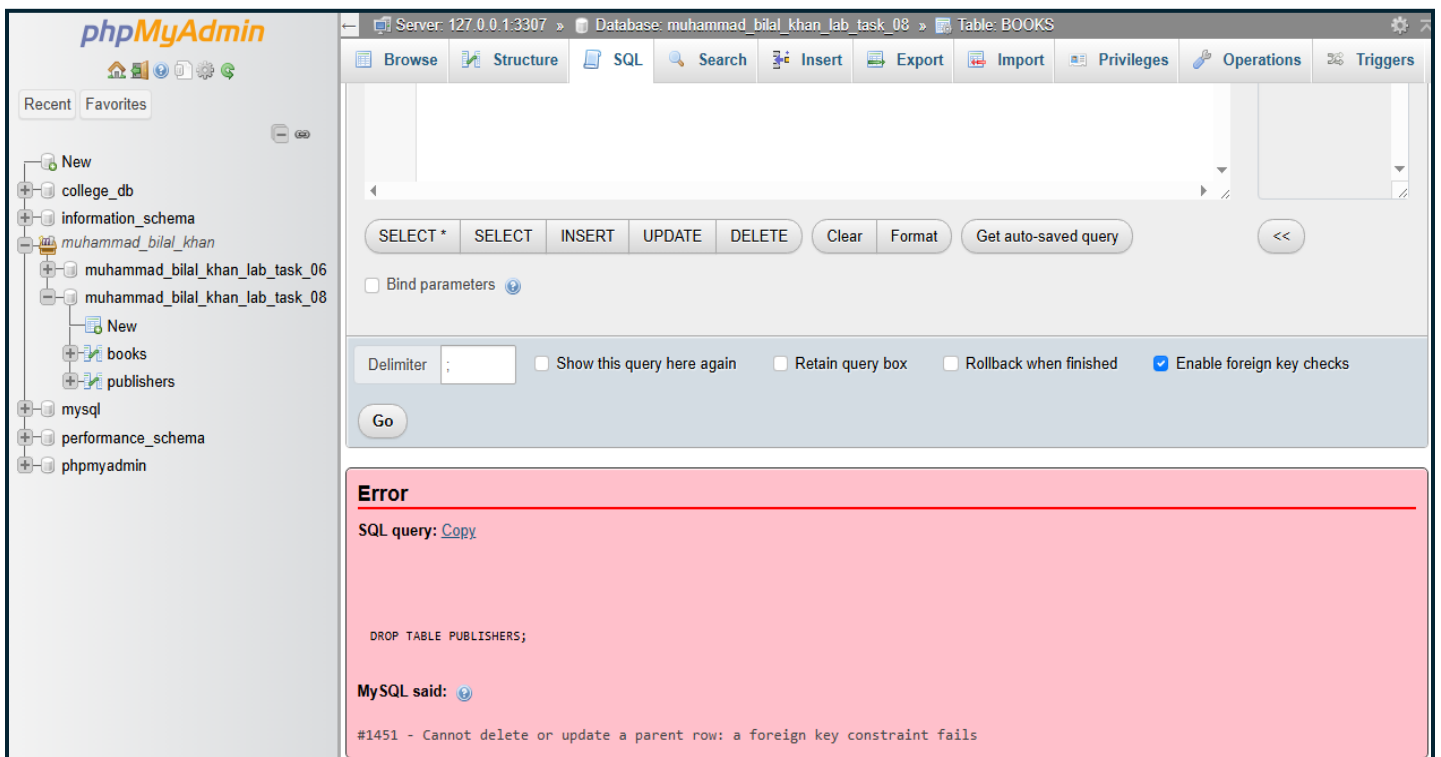
Section 05:

Managing Foreign Key Dependencies for Table Deletion

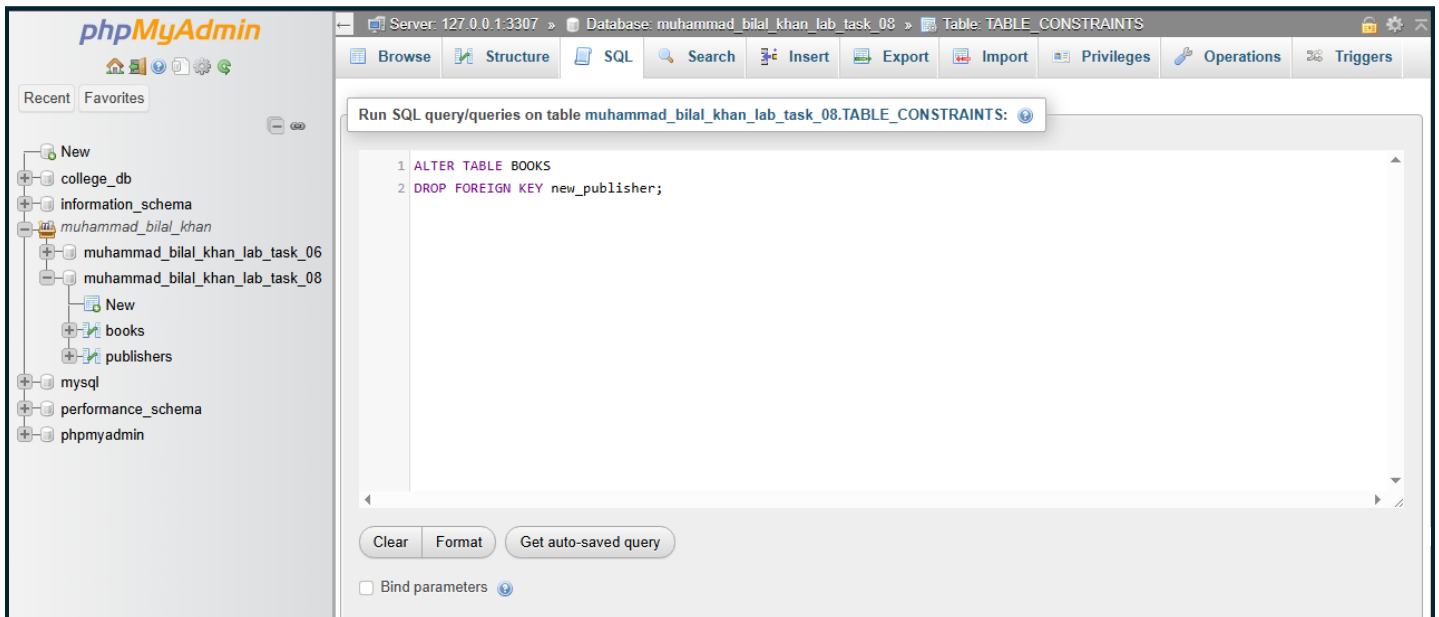
1. Attempt to delete the PUBLISHERS table using DROP TABLE.



2. Observe the error message related to the foreign key dependency.



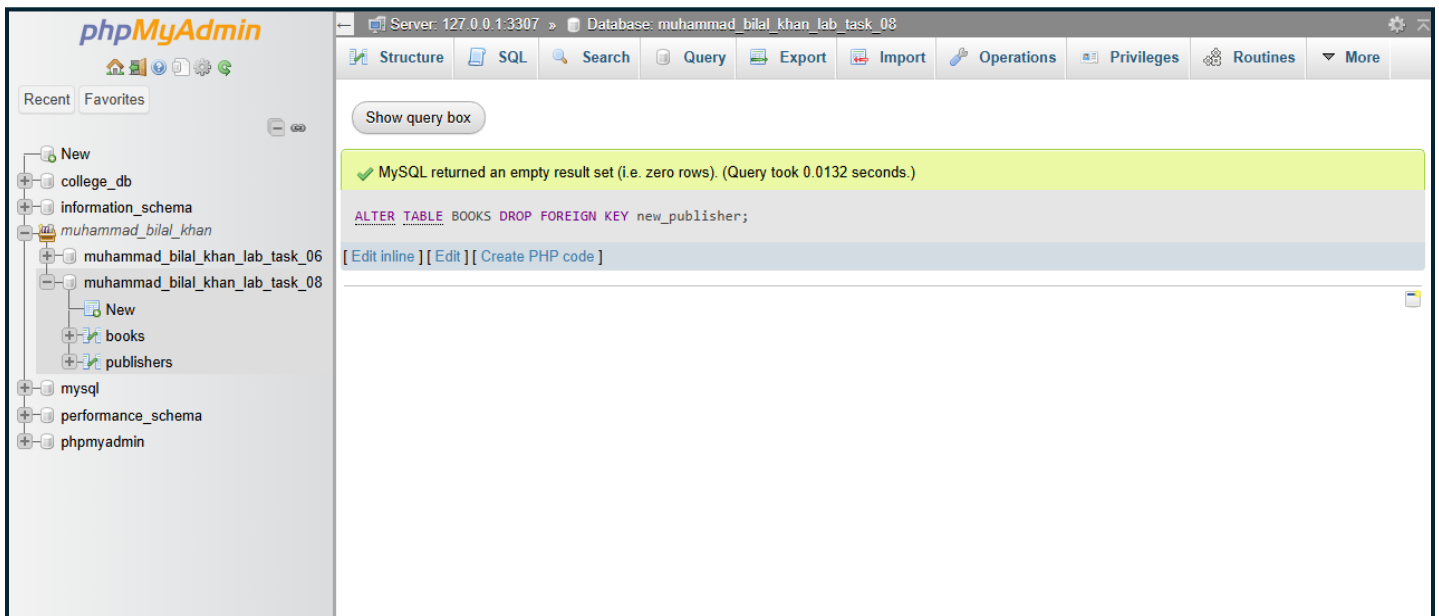
3. Use ALTER TABLE to drop the foreign key constraint from the BOOKS table.



The screenshot shows the phpMyAdmin interface. On the left, the database structure is visible, including the 'muhammad_bilal_khan_lab_task_08' database with tables 'books' and 'publishers'. The main panel is in the 'SQL' tab, displaying the following query:

```
1 ALTER TABLE BOOKS
2 DROP FOREIGN KEY new_publisher;
```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query', along with a checkbox for 'Bind parameters'.



The screenshot shows the phpMyAdmin interface after the query has been executed. The main panel displays a success message: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0132 seconds.)'. Below the message, the executed query is shown: 'ALTER TABLE BOOKS DROP FOREIGN KEY new_publisher;'. There are links for '[Edit inline]', '[Edit]', and '[Create PHP code]'.

4. After removing the foreign key, successfully delete the PUBLISHERS table.

